```
5000 ' Subroutine to determine if the pixel at X%[R%,C%] can be removed.
5010 ' If all four of the rules are satisfied, then X%(R%,C%], is set to a value of 1,
5020 ' indicating it should be removed at the end of the iteration.
5030 '
5040 'RULE #1:  Do nothing if the pixel already white
5050 IF X%(R%,C%) = 255 THEN RETURN
5060 '
5070 '
5080 'RULE #2:  Do nothing if all of the close neighbors are black
5090 IF  X%[R% -1,C%  ] <> 255 AND X%[R%  ,C%+1] <> 255 AND
          X%[R%+1,C%  ] <> 255 AND X%[R%  ,C% -1] <> 255  THEN RETURN
5100 '
5110 '
5120 'RULE #3:  Do nothing if only a single neighbor pixel is black
5130 COUNT% = 0
5140 IF X%[R% -1,C% -1]      = 0 THEN COUNT% = COUNT% + 1
5150 IF X%[R% -1,C%    ]     = 0 THEN COUNT% = COUNT% + 1
5160 IF X%[R% -1,C%+1]  = 0 THEN COUNT% = COUNT% + 1
5170 IF X%[R%    ,C%+1]  = 0 THEN COUNT% = COUNT% + 1
5180 IF X%[R%+1,C%+1]  = 0 THEN COUNT% = COUNT% + 1
5190 IF X%[R%+1,C%    ]     = 0 THEN COUNT% = COUNT% + 1
5200 IF X%[R%+1,C% -1]      = 0 THEN COUNT% = COUNT% + 1
5210 IF X%[R%    ,C% -1]     = 0 THEN COUNT% = COUNT% + 1
5220 IF COUNT% = 1 THEN RETURN
5230 '
5240 '
5250 'RULE 4:  Do nothing if the neighbors are unconnected.
5260 'Determine this by counting the black-to-white transitions
5270 'while moving clockwise through the 8 neighboring pixels.
5280 COUNT% = 0
5290 IF X%[R% -1,C% -1] = 0 AND X%[R% -1,C%    ]          > 0  THEN COUNT% = COUNT% + 1
5300 IF X%[R% -1,C%    ] = 0 AND X%[R% -1,C%+1]          > 0  AND X%[R%  ,C%+1] > 0
          THEN COUNT% = COUNT% + 1
5310 IF X%[R% -1,C%+1]      = 0 AND X%[R%    ,C%+1]     > 0  THEN COUNT% = COUNT% + 1
5320 IF X%[R%    ,C%+1]     = 0 AND X%[R%+1,C%+1]      > 0  AND X%[R%+1,C%  ] > 0
          THEN COUNT% = COUNT% + 1
5330 IF X%[R%+1,C%+1]      = 0 AND X%[R%+1,C%    ]     > 0  THEN COUNT% = COUNT% + 1
5340 IF X%[R%+1,C%    ]     = 0 AND X%[R%+1,C% -1]     > 0  AND X%[R%  ,C%-1] > 0
          THEN COUNT% = COUNT% + 1
5350 IF X%[R%+1,C% -1]      = 0 AND X%[R%    ,C% -1]     > 0  THEN COUNT% = COUNT% + 1
5360 IF X%[R%    ,C% -1]     = 0 AND X%[R% -1,C% -1]     > 0  AND X%[R%-1,C%  ] > 0
          THEN COUNT% = COUNT% + 1
5370 IF COUNT% > 1 THEN RETURN
5380 '
5390 '
5400 'If all rules are satisfied, mark the pixel to be set to white at the end of the iteration
5410 X%(R%,C%) = 1
5420 '
5430 RETURN
```

TABLE 25-2

by having only one black neighbor. Do this several times and the fuzz is removed at the expense of making each of the correct lines shorter. A better method would loop through the image identifying *branch pixels* (pixels that have more than two neighbors). Starting with each branch pixel, count the number of pixels in each offshoot. If the number of pixels in an offshoot is less than some value (say, 5), declare it to be fuzz, and change the pixels in the branch from black to white.